

SECTION A

1 (**long**) The FET amplifier shown in Fig. 1 is to be biased so that $V_{GS} = -2$ V, $V_{DS} = +8$ V and $I_D = 4$ mA. At this operating point, the FET has small-signal parameters of $g_m = 5$ mA V⁻¹ and $r_d = 10$ k Ω .

- Explain how the resistor chain on the left of the FET sets the operating point of the FET. [4]
- Determine the values of R_1 and R_2 to achieve the desired operating point. [4]
- Draw a small-signal equivalent circuit for the amplifier circuit of Fig. 1, valid for mid-band frequencies. Hence determine the small-signal gain and the output impedance of the circuit. [10]
- The low frequency half-power frequency of the amplifier is dominated by the effect of C_0 , and is designed to be 10 Hz. With a load resistor of 5 k Ω connected between points A and B, determine the value of C_0 . [8]
- Given the choice of gate-source voltage of -2 V, explain why $V_{DS} = 12$ V is the optimum choice of the drain-source voltage. [4]

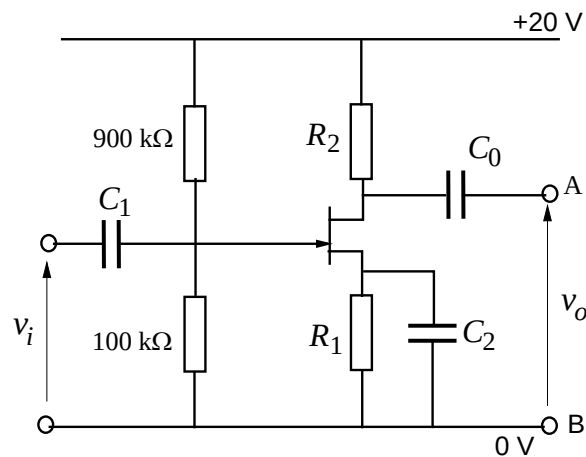


Fig. 1